

Consequences of Ambivalence on Satisfaction and Loyalty

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ABSTRACT

The key objective of this study was to understand the consequences of subjective ambivalence on customer satisfaction, loyalty, and the satisfaction–loyalty relationship. The conceptual and theoretical discussions were derived largely from recent research in social psychology and integrated with marketing literature on satisfaction and loyalty. Given that product evaluations are typically positive and extreme, these findings indicate a negative relationship between ambivalence and satisfaction. Even though a great deal of the variance in ambivalence is shared with satisfaction, ambivalence did prove to have an independent and direct effect on loyalty. Ambivalent consumers are not only less loyal because they are less satisfied, but for other reasons, as well. Ambivalence was not found to moderate the satisfaction–loyalty relationship. The results of the study underscore the importance of taking ambivalence into consideration when measuring satisfaction and modeling satisfaction–loyalty relationships. © 2005 Wiley Periodicals, Inc.

Consider a typical consumption situation in which a couple is having dinner at a restaurant. The atmosphere and décor are superb, but they had to wait over 40 minutes before receiving their meals. The wine selection was surprisingly good, the seafood was overcooked, the salad ingredients were fresh but there was too little salad dressing. The dessert was creative and delivered promptly, while the two servers they had during their stay were very different in terms of reliability and helpfulness. The obvious question is, did these customers leave the restaurant feeling satisfied with their experience, dissatisfied with their experience, or both?

Many studies in the consumer literature have discussed different aspects of mixed emotions or feelings (e.g., Ruth, Brunel, & Otnes, 2002). But, in spite of the fact that Otnes, Lowrey, and Shrum (1997) noted the potential importance of ambivalence in consumer research, the present authors are not aware of any studies that have empirically addressed the presence or the impact of ambivalence in consumer-satisfaction studies. Sometimes when consumers express their evaluations of products and services with positive or negative ratings of satisfaction, joy, happiness, likeability, or delight, they express no single dominant evaluative response but rather two strongly conflicting evaluations, each based on a consistent set of cognitions or feelings (Kaplan, 1972; Lavine, Thomsen, Zanna, & Borgida, 1998). Standard rating-scale formats such as those provided by Likert scales or semantic-differential scales do not allow for the *unique* expression of evaluative ambivalence. Instead, instructing respondents to put only one mark on the scale may very well force them to give answers that reflect their attempts at finding a compromise between positive and negative evaluations. This makes it difficult to interpret the individual responses unambiguously. For example, choice of the middle category on an attitude scale may be due to a nonambivalent neutral evaluation (indifference), nonattitudes, or to ambivalence (Klopfer & Madden, 1980; Olsen, 1999).

Although ambivalence may be underexplored in consumer research, it has recently received much attention in other disciplines, such as political science, psychology, and social psychology (Jonas, Broemer, & Diehl, 2000a). Ambivalence has been defined in various ways across these disciplines, but most definitions incorporate mixed or conflicted feelings associated with an attitude object (Conner & Sparks, 2002). Ambivalent attitudes seem to be less extreme than unambivalent ones (Brömer, 1998; Klopfer & Madden, 1980). Ambivalence has so far been studied as a strength-related moderator between attitude and behavior (Jonas, Broemer, & Diehl, 2000b). Some studies have shown that ambivalent attitudes lead to a decrease in the magnitude of the attitude → intention or attitude → behavior relationship (Armitage & Conner, 2000; Sparks, Hedderley, & Shepherd, 1992), whereas others suggest that attitude ambivalence actually strengthens the relationship between attitude and intention as well as between attitude and behavior (Jonas, Diehl, & Brömer, 1997).

Based upon the significant role ambivalent attitudes have played in other literatures, and given the fact that satisfaction has been hypothesized as one of the more important antecedents of repurchase loyalty (Ganesh, Arnold, & Reynolds, 2000; Hennig Thureau & Klee, 1997; Oliver, 1999), it seems logical to explore the impact of ambivalence on satisfaction, loyalty, and the relationship between the two. An added motivation is that there have been relatively few empirical examinations of the satisfaction–loyalty relationship itself (Bloemer & de Ruyter, 1998; Mittal & Kamakura, 2001; Zeithaml, 2000), and what moderates this relationship (Holmburg & Giering, 2001). Thus, the research reported below seeks to address three questions: Is ambivalence, when present, a predictor of satisfaction? Is ambivalence a predictor of repurchase loyalty? Does ambivalence influence the satisfaction–loyalty relationship and if so, how?

In order to address these questions, the literature on attitudinal ambivalence is reviewed and then integrated into the satisfaction literature. Second, hypotheses are generated from this integration, which are then tested by means of survey data. Although many of the studies on ambivalence have utilized experiments with student subjects, the present investigation uses data from a sample of 1,200 Norwegian consumers of seafood. The results of a nationwide survey may or may not be the same as those from a controlled or survey experiment on student populations (Armitage & Connor, 2000). Finally, implications of the findings are discussed and future research directions are provided.

THEORETICAL FOUNDATIONS AND HYPOTHESES

Research on attitude ambivalence has suggested that conflicting and inconsistent feelings may underlie the evaluation of a single object or that attitude evaluations are based on separate positive and negative evaluations of different attitude object components (Thompson, Zanna, & Griffin, 1995). For example, Gardner (1987) describes ambivalence as a “psychological state in which a person holds mixed feelings (positive and negative) towards some psychological object” (p. 241). Although attitudinal ambivalence is actually a reconceptualization of the attitude construct, it is more commonly treated as a part of attitude strength (Olsen, 1999; Thompson et al., 1995). Because of their dualistic evaluative structure, ambivalent attitudes are presumed to be weaker than attitudes based on coherently positive or negative cognitions or feelings (Brömer, 1998).

Krosnick and Petty (1995) argue that there are four defining features of strong attitudes. They (a) remain stable over time, (b) impact behavior, (c) influence information processing, and (d) resist persuasion. In addition, strong attitudes are believed to be more accessible in memory, more extreme in their expressed evaluation, and held with more confidence and knowledge. Therefore, it might be expected that ambivalent

attitudes will be less stable over time, less predictive of behavior, more pliable, less extreme, less accessible in memory, and held with less confidence. A number of authors have investigated the effect of attitudinal ambivalence on each of the Krosnick and Petty (1995) defining features but have produced conflicting findings (see Conner & Sparks, 2002, for a recent review). Because this study focuses on how ambivalence influences satisfaction, loyalty, and the satisfaction–loyalty relationship, the literature review will concentrate on attitude studies that can be used to understand the relationship between these constructs.

Ambivalence, Attitude, and Extremity

Attitude and evaluative responses in attitude research are defined by their valence and extremity. The valence component deals with the direction of the evaluation (Eagly & Chaiken, 1993), whereas extremity involves how strongly that position is held (Zanna & Rempel, 1988). Attitude extremity has typically been operationalized as the deviation of an individual's attitude rating from the midpoint of a self-report rating scale (Judd & Brauer, 1995). According to Fazio (1995), individuals with differentially polarized attitudes (or extreme attitudes) actually hold different attitudes rather than holding the same attitude with different degrees of strength. Thus, extremity reflects not a quality of the attitudinal representation, but the attitude *per se*.

Early studies on ambivalence (Kaplan, 1972) attempted to distinguish ambivalence from indifference by focusing on responses using the middlemost category on the attitude scale. Other efforts suggested that ambivalent respondents tended to use the mid-point responses more often than unambivalent respondents (DuBois & Burns, 1975; Klopfer & Madden, 1980). Recent studies of the relationship between ambivalence and attitude extremity provide mixed findings. Brömer (1998) found that three different measures of ambivalence were negatively correlated with attitude extremity. In another study by Jonas, Broemer, and Diehl (2000b) the relationship between perceived ambivalence and attitude extremity was reported to be nonsignificant.

More recent social psychological work views attitudes as spontaneous constructions that are derived whenever there is some need to form or express evaluations (Wilson & Hodges, 1992). Viewing attitudes as temporary or spontaneous constructions, however, does not mean that no prior evaluation exists: It refers to the notion that attitudes are a function of activated prior evaluations and currently accessible cognitions or feelings about some attitude object. When two opposing evaluative tendencies exist, individuals should be inclined or even forced to integrate any prior evaluation with current assessments into a single evaluative response. Thus, due to inherent internal inconsistency, ambivalent attitudes are likely to be less extreme (Thompson, Zanna, & Griffin, 1995). Based on these assumptions and some empirical findings (e.g., Brömer,

1998; Klopfer & Madden, 1980), this study hypothesizes that ambivalence is reflected in the simultaneous existence of both negative and positive satisfaction of products, and is negatively associated with satisfaction extremity.

Ambivalence, Intention, and Behavior

Because ambivalence has been studied as a property of attitudes and as a moderator variable (Thompson et al., 1995), the direct relationship between ambivalence and behavior has not been a focal issue. However, several other strength-related properties of attitude, such as amount of product knowledge and degree of product involvement, are assumed to be positively related to behavior (Eagly & Chaiken, 1993). It also follows from the assumptions in the previous paragraphs that if univalent attitudes are more extreme than ambivalent attitudes, they therefore have the potential to be more predictive of behavioral intentions and behavior itself. This hypothesis is consistent with recent studies (e.g., Lavine et al., 1998).

Although *actual* behavior is seldom reported, two recent studies have reported on the relationship between ambivalence and behavioral intentions. In Sparks et al. (2001), the relationship between ambivalence and the intention to consume chocolate was negative, but not significant, whereas the relationship between ambivalence and the intention to consume meat was negative and significant. In this study, the correlation between attitude and intention was high (e.g., .71 for meat consumption).

In a study based on the theory of planned behavior, Povey, Wellens, and Conner (2001) found ambivalence to be a significant predictor of intention to eat meat, after attitude, subjective norm, perceived behavioral control, and self-identity were entered into the regression equation. In all of the studies reviewed, the relationship between ambivalence and intention was negative, although not always significant. Based on these results and those involving other strength-related properties (Lavine, Borgida, & Sullivan, 2000), ambivalence is hypothesized to be negatively related to loyalty or consumption frequency.

Ambivalence and the Attitude–Behavior Relationship

The influence of ambivalence on the attitude–intention relationship has been investigated in a handful of studies. According to Sparks, Hedderley, and Shepherd (1992), ambivalent attitudes are based upon discrepant evaluations and should for that reason lead to a decrease in attitude–intention consistency. The results from their original study and a follow-up study by some of the same authors (Sparks, Conner, James, Shepherd, & Povey, 2001) confirmed this hypothesis. Two other studies (Moore, 1980; Povey et al., 2001) report results consistent with the

assumption that ambivalent attitudes are associated with lower attitude–intention consistency.

Although these studies focused on the moderating role of ambivalence on attitude–intention consistency, some recent published and unpublished studies demonstrate an analogous effect on the *attitude–behavior* relationship (see Conner & Sparks, 2002, for a review). Armitage and Conner (2000) have shown that ambivalence is related to lower attitude–behavior correlation. They found that ambivalence moderated both the attitude–behavior and the attitude–intention relationships. Univalent attitudes were more predictive of both intention and behavior than were ambivalent attitudes; however, greater attitudinal ambivalence was associated with a stronger intention–behavior relationship. Their study also suggested that less ambivalent attitudes predicted behavior directly, whereas more ambivalent attitudes exerted an indirect influence on behavior via behavioral intention.

Thus, findings relating to the impact of ambivalence are inconsistent across studies. Jonas et al. (2000b) found that ambivalence moderated the relationship between attitude and actual behavior for only one out of six behavioral phenomena they examined in their experiment (active participation in class). Discovering multicollinearity between the independent variable and the multiplicative term as suggested by Aiken and West (1991), Jonas et al. reduced multicollinearity by subject-centering both the independent (attitude) and the moderator (ambivalence) variables. Resulting correlations ranged from .01 to $-.62$, and were lowest when ambivalence was found to be a moderator of the attitude–behavior relationship (active participation in class). In a second study, they reported that moderating effects of ambivalence were only observed when multicollinearity between the independent variables and the interaction terms was very low.

Thus, it may be that ambivalence does not always act as a moderating variable (Baron & Kenny, 1986). It could serve as a quasi-moderator (Sharma, Durand, & Gur-Arie, 1981), or in some cases may simply be a predictor of attitude/satisfaction and behavior/loyalty. Whether or not ambivalence will behave as a moderator in this particular study is, thus, an open question (Fournier & Mick, 1999; Mittal & Kamakura, 2001). However, based on some of the findings within social psychology, it might be expected that ambivalence, if found to be a moderator, will weaken the relationship between satisfaction and loyalty.

Ambivalence within a Satisfaction–Loyalty Framework

Satisfaction has been defined and operationalized in various ways (Yi, 1990). One of the more recent definitions of satisfaction as a composite construct was developed by Oliver (1997) who proposed it to be “the consumer’s fulfillment response, the degree to which the level of fulfillment is pleasant or unpleasant” (p. 28). Although different approaches are used to measure satisfaction, respondents are requested to put one and

only one mark on a scale ranging from a negative to a positive evaluation. This approach is in accordance with the common view of studying attitudes as a summary evaluation of objects along a bipolar, favorable–unfavorable continuum (Eagly & Chaiken, 1993).

As noted above, a few studies have discussed ambivalence within the consumer-behavior literature. Otnes et al. (1997) defined consumer ambivalence as “the simultaneous or sequential experience of multiple emotional states, as a result of the interaction between internal factors and external objects, people, institutions, and/or cultural phenomena in market-oriented contexts, that can have direct and/or indirect ramifications on prepurchase, purchase, or postpurchase attitudes and behavior” (pp. 82, 83). Their broad definition is based on a variety of psychological, sociological, and cultural perspectives on ambivalence. It suggests that ambivalence can result from the interplay between internal emotional states and expectations, and external components that are specific to the marketplace. Thus, within this definition, ambivalence can play a role as a predictor, moderator, and mediator to attitude and behavior.

The discussion of different modes of satisfaction is not new. Several studies over the last few years deal with a multicomponent approach in studying the satisfaction response. Theoretical constructs of evaluation, separate negative and positive affect dimensions, and emotion and satisfaction response are studied within different conceptual frameworks (Mano & Oliver, 1993; Oliver, 1997). Westbrook and Oliver (1991) identified three dimensions underlying satisfaction: hostility (negative affect), pleasant surprise (positive affect), and interest (something that might be termed importance or involvement). Fournier and Mick (1999) provide a more holistic view of consumer satisfaction. They suggest that satisfaction is an active, dynamic process; it contains a strong social dimension; and it incorporates meaning and emotions as integral components. Within these perspectives, ambivalence can be viewed both as an antecedent of satisfaction (Mano & Oliver, 1993) and as a predictor of consumption (Arnould, Price, & Zinkhan, 2002).

Loyalty has been defined and measured in many different ways. The most widely agreed-upon definition of loyalty is “behavioral response expressed over time by some decision-making unit with respect to one or more alternatives” (Dick & Basu, 1994; Jacoby & Chestnut, 1978). Oliver (1997) defines loyalty as “a deeply held commitment to rebuy or repatronize a preferred product or service in the future” (p. 392). Several authors tend to consider loyalty on a multidimensional basis, by adding an attitudinal (liking or preferences), motivational, or conative (intention or commitment to consume) component to a behavioral loyalty concept (Chaudhuri & Holbrook, 2001; Ganesh et al., 2000; Yi & Jeon, 2003). Combinations of past frequent behavior and intention to rebuy (e.g., Nijssen, Singh, Sirdeshmukh, & Holzmüller, 2003; Pritchard, Havitz, & Howard, 1999) are also used to assess a global and cumulative loyalty measure.

Loyalty is considered here as a cumulative construct including both the act of consuming or reconsumption (action loyalty) and intended or

expected consumption (conative loyalty or future rebuying) in terms of Oliver's (Oliver, 1997, 1999) recent loyalty framework. One of the advantages of assessing loyalty as a cumulative construct is to avoid problems with causal reasoning in cross-sectional data. Most studies on ambivalence use a prospective design and measure behavior some weeks or months after attitudes and intentions (e.g., Armitage & Conner, 2000), but studies on satisfaction–loyalty relationships use cross-sectional data and suggest that satisfaction causes loyalty (Dick & Basu, 1994; Oliver, 1999).

The relationships between customer satisfaction and intention (e.g., Anderson & Sullivan, 1993; Taylor & Baker, 1994) are assumed to be positive, but vary between products, industries, and situations (Cronin, Brady, & Hult, 2000; Fornell, Johnson, Anderson, Cha, & Bryang, 1996). Despite the extensive discussion and important consequences of the satisfaction, few empirical studies have tested their relationship to actual, reported buying behavior or loyalty (Hennig Thureau & Klee, 1997; Zeithaml, 2000), but the indication is the same as for intention—a positive relationship (Bloemer & de Ruyter, 1998). A recent study by Mittal and Kamakura (2001) indicates that the relationship between satisfaction and repurchase behavior is complicated, can result in no correlation, and may be moderated by several customer characteristics. Their view supports some findings that satisfaction is related to several internal, social, external, and contextual variables (Fournier & Mick, 1999).

Research Issue

Despite the growing interest in ambivalence within attitude research and measurement, formal attention by means of conceptual or empirical studies has been slow in coming (Conner & Sparks, 2002). The study reported below seeks to add to the literature by addressing the three research questions presented in the introduction: (1) Is ambivalence, when present, a predictor of satisfaction? (2) Is ambivalence a predictor of repurchase loyalty? (3) Does ambivalence influence the satisfaction–loyalty relationship, and if so, how?

The first question deals with ambivalence and the extremity of satisfaction responses. Consumers for the most part are satisfied with products they consume and therefore are likely to report high scores on a satisfaction scale (Peterson & Wilson, 1992). Ambivalent consumers, however, are expected to be less satisfied than unambivalent consumers. Thus, based on earlier studies of midpoint responses on attitude scales (Klopfer & Madden, 1980), and more recent studies of ambivalence (Lavine et al., 1998), a negative relationship between ambivalence and satisfaction (extremity) is expected. The second issue involves the relationship between ambivalence and loyalty. If, as expected, more ambivalent consumers are less satisfied, and less satisfied consumers tend to be less loyal, ambivalent consumers should be less loyal (lower intention and frequency of buying). Based on this logic and research available

(Povey et al., 2001; Sparks et al., 2001), a negative relationship between ambivalence and loyalty is expected. This may also imply that ambivalence is not a pure moderator, but what Sharma et al. (1981) term a “quasi moderator.”

The third issue of this study addresses the effect of ambivalence on the satisfaction–loyalty relationship, an area where conflicting results are present (e.g., Jonas et al., 1997). In studies involving food as the attitude object (Conner & Sparks, 2002; Sparks et al., 2001) ambivalence was found to act as a moderator. However, according to Jonas et al. (2000b) the moderating effect itself may be dependent on the situation and possibly the multicollinearity between the independent variable (satisfaction) and the interaction term. Recent research within consumer behavior indicates that the relationship between satisfaction and repurchase loyalty is more complicated than suggested earlier (Mittal & Kamakura, 2001). However, it is expected that ambivalence will be a quasimoderator that to some extent may weaken the relationship between satisfaction and loyalty.

RESEARCH METHOD

Research Setting and Subjects

The data used in this study come from a large, representative national survey of attitudes toward and consumption of seafood by Norwegian consumers. The analysis is based on 1,194 respondents answering all of the questions dealing with satisfaction, ambivalence, and loyalty. The questionnaire was addressed to the person responsible for preparing the family dinner. The average age of the respondents was 43 years, but the study covered all age groups from 20 to 70.

Seafood was felt to be an appropriate attitude object and Norwegians appropriate subjects in this study for three reasons. First, food is an object about which mixed feelings may be held (Armitage & Conner, 2000; Letarte, Dubé, & Troche, 1997). Food evaluation and choice is often used as an example to illustrate people’s internal conflict, ambivalence, competing motives, changing preferences, or their “weakness of will” (Beardsworth, 1995; Sparks et al., 2001). Second, food preparation and consumption reflects years of experience on the part of these respondents, avoiding the problem presented by newly formed attitudes (Armitage & Conner, 2000; Fazio, 1995). Finally, seafood is a mainstay of Norwegian diets, ensuring a high incidence rate of familiarity with the product.

Questionnaire and Measurement of the Constructs

Question order effect is a well-documented phenomenon in survey research (Bicart, 1993). Recent research on question-order effect in sat-

isfaction–loyalty modeling suggests that the impact of satisfaction on loyalty is quite robust to order effects in a product/service context and when satisfaction is measured as a cumulative evaluation (Auh, Salisbury, & Johnson, 2003). It is believed that questions about behavioral frequency are more demanding and difficult to respond to than those involving attitude evaluation (Tourangeau, Rips, & Rasinski, 2000), and responses to early items should be more accurate than responses to later items (Sudman, Bradburn, & Schwarz, 1996). The questionnaire used in this study was organized to attempt to minimize these effects. The first page of the questionnaire addressed consumption frequency and intention to consume. Questions about preferences for different forms of seafood (not used in this study) were next, followed by questions concerning satisfaction and ambivalence. The insertion of preference questions between questions on loyalty and satisfaction should reduce carryover effects, as well. The measures employed are reported below.

Satisfaction. Satisfaction is defined as a global evaluation or feeling state. Oliver's (1997) recent definition of satisfaction was used as the basis for the satisfaction measure. The respondents were asked to express their agreement or disagreement by two statements of general evaluation: "I feel very *satisfied* when I have seafood for dinner," and "Seafood for dinner gives me a *pleasant feeling*." These two items also cover general positive feeling statements often used to assess attitudes toward food objects and/or food behavior (Sparks et al., 1992; Rozin, 1995).

Repurchase Loyalty. The most common assessments of loyalty are behavioral frequency or repurchase patterns (Bloemer & Kasper, 1995; Jacoby & Chestnut, 1978). A combination of repurchase behavior and intention to purchase has been used as a cumulative measure of loyalty (e.g., Nijssen et al., 2003; Pritchard, Havitz, & Howard, 1999), and a similar approach was taken here (Fornell et al., 1996; Ganesh et al., 2000). The two measures of self-report consumption covered both general frequency and recent frequency. These measures are analogous with several studies testing frequent food behavior using expectancy–value attitude models (e.g., Raats, Shepherd, & Sparks, 1995). The general frequency measures of behavior used a 1-year time frame and were addressed by a 9-point scale of the form "How many times—on average—during the last year have you eaten seafood for dinner in your home? 9 = 3 times a week, 8 = twice a week, 7 = once a week, 6 = once every 2 weeks, 5 = once a month, 4 = 1–2 times each quarter, 3 = 1–2 times every six months, 2 = more seldom, 1 = never. Recent frequency was assessed by a 15-point scale of the form "Can you estimate how many times during the last 14 days you have eaten at least one form of seafood (cod, salmon, herring, coalfish, halibut . . .) or seafood products (fishcakes . . .) for dinner in your home: 0, 1, 2, . . . , 14 times or more." The intention measure was framed as a behavioral expect-

tation (Warshaw & Davis, 1985) and assessed by asking the respondents, "How many times in the coming 14 days do you expect to eat seafood and seafood products for dinner?" The scale had 15 possibilities ranging from 0 to 14 times or more.

It should be noted that measurement in combination with location of the loyalty construct appears to violate the conditions for demonstrating causality. The construct is comprised of two measures: one indicating behavior over the past year, the other over the past 2 weeks. Moreover, the study predicts past behavior with independent variables that do not explicitly contain mention of a time frame, but presumably refer to how respondents felt at the moment when the items were filled out. Thus, in actuality, a composite of past loyalties over two different time frames are predicted by present judgments of subjective ambivalence and satisfaction referring to a period in time after loyalty outcomes occurred. This would, of course, be a fatal flaw were the study conducted with longitudinal data but, needless to say, such was not the case in the current study.

Subjective Ambivalence. Previous research into attitudinal ambivalence has operationalized ambivalence in several ways. One approach is to assess the subjective perception of ambivalence *directly*, by asking individuals, for example, whether their attitudes are ambivalent, mixed, or in conflict toward an attitude object. This approach focuses on the psychological and perceived experiences with ambivalence and conflict associated with an attitude (Lavine et al., 2000). People who respond that their feelings are ambivalent, inconsistent, rapidly interchangeable, mixed, or in conflict are categorized as holding more ambivalent attitudes than those who respond that their feelings are one-sided (Conner & Sparks, 2002; Sparks et al., 1992). This approach is termed *subjective ambivalence*.

The second method of assessing ambivalence is to have individuals provide separate ratings of their positive and negative reactions to an attitude object. These positive and negative reactions are then combined according to mathematical models to provide an index of ambivalence elicited by the attitude object (Breckler, 1994). The formula-based measure of ambivalence is often termed *objective ambivalence*. At least six models of how to combine the positive and negative bases of attitudes to form an objective ambivalence index have been offered (Priester & Petty, 1996).

Several authors tend to view measures of subjective ambivalence as the gold standard for the validation of formula-based measures (e.g., Thompson et al., 1995). However, subjective ambivalence assumes that respondents are aware of their ambivalence or have conscious access to their ambivalence (Jonas et al., 2000b). In contrast, formula-based measures do not assume that respondents are aware of their degree of ambivalence. In that respect, both methods are acceptable as different approaches to measure ambivalence. The correlation between subjective and objec-

tive measures of ambivalence is reported to be in the range of .21–.52 (Priester & Petty, 1996; Thompson et al., 1995).

To assess the degree of subjective ambivalence, three items were employed. Respondents answered by marking the appropriate position on the 7-point Likert scale framed as “Sometimes I feel satisfied, but other times I feel dissatisfied when eating seafood,” “Sometimes I feel seafood tastes good, but other times I feel it tastes bad,” and “Sometimes I feel that seafood is exciting, but other times I feel it boring.” This scale is consistent with the way Klopfer and Madden (1980) measured ambivalence: “Sometimes you feel one way and sometimes you feel another.” It is also in accordance with one of the three approaches Gardner (1987) suggests to measure ambivalence. The subjective ambivalence scale used in this study correlated .52 ($p < .001$) with an objective measure of ambivalence calculated from different positive and negative attribute evaluations with the use of the Griffin calculation (Sparks et al., 2001; Thompson et al., 1995).¹ A pretest of the questionnaire using scale items, including words such as “mixed feelings,” “ambivalence,” “indecision,” “mixed reactions,” or “feel conflict,” proved to be more difficult to comprehend and provided less reliable results. In a free-elicitation exercise, the most common verbal expression when having mixed feeling about eating seafood for dinner was “Both—and” as a spontaneous response, followed by different variations of “Sometimes it tastes good, and sometimes I don’t like it.”

Analytical Procedures

Three separate approaches were required to examine the research questions. First, measures of the constructs were evaluated with the use of confirmatory factor analysis in order to examine discriminant validity and construct reliability. Next, the structural relationships between ambivalence and satisfaction and between ambivalence and loyalty were examined simultaneously by means of structural equation modeling. These analyses were conducted with the use of maximum-likelihood estimation in LISREL 8.50 (Jöreskog & Sörbom, 1993), based on covariances. The traditional χ^2 (chi square) fit test is reported, however, because it has been recognized as an inappropriate test for large sample size

¹ Objective ambivalence for each respondent was calculated by using a formula developed by Thompson et al. (1995): $(P + N)/2 - |P - N|$, where P is the sum of evaluations on positive dimensions and N is the sum of negative evaluations. This formula is used in several recent studies measuring food ambivalence (e.g., Sparks et al., 2001). The positive statements included four items: “Seafood tastes good,” “It is healthier to eat seafood than chicken,” “It is wise to have seafood for dinner,” and “I feel it is convenient to prepare a seafood dinner.” The composite score of negative evaluations was derived from another four items: “Seafood is boring as a dinner,” “Fish bones spoil the pleasure of eating seafood,” “The smell of fish spoils the pleasure of eating seafood,” and “I feel it is disgusting to prepare seafood for dinner.” In terms of empirical comparisons, Thompson et al. (1995) suggest that the above method of calculating objective ambivalence resulted in higher correlations with an independent measure of ambivalence than did the other methods.

(Browne & Cudeck, 1992), three other indices are also included: root-mean-square error of approximation (RMSEA), nonnormed fit index (NNFI), and comparative-fit index (CFI). Acceptable model fits are indicated by NNFI and CFI values exceeding 0.90, and RMSEA values below 0.08 represent a moderate fit, whereas values less than 0.05 are considered good (Browne & Cudeck, 1992).

Examination of the possible effect of ambivalence on the *relationship* between satisfaction and loyalty required a different approach, however. The usual procedure to examine the effect of a moderator variable in structural-equation models (SEMs) involves dividing the sample into groups formed by a median split on the possible moderator variable. Identical models are estimated within each of the groups and compared. If the structural coefficients differ significantly, there is evidence of a moderating effect. If the moderator is also a predictor, however, the median split approach should not be used. The reduction in variance in the predictor caused by the split will be present in the dependent measure as well, because of the relationship between the two (Peters & Champoux, 1979). Effects for representing and testing interaction effects are familiar in regression (Cohen & Cohen, 1983; Sharma et al., 1981). Recent developments by Jöreskog and Yang (1996) and Jonsson (1998) have made it easier to apply the Kenny and Judd (1984) indicant product analysis procedure in structural equation modeling. The procedure involves substantial complexity in the model specification and large sample sizes in order to provide valid test statistics.

RESULTS

Validations of Measures

Apart from the χ^2 statistics (31.45; 17 *df*, $p = .018$), the measurement model of the three main constructs exhibited good fit (RMSEA = 0.027; NNFI = 0.99; and CFI = 1.00). Convergent validity was examined by looking at the individual item loadings on the constructs and the average measure of variance shared between the items and the construct (Jöreskog & Sörbom, 1993). Reliability of the multi-item scales was assessed by computing Jöreskog's reliability coefficient for each construct (J. C. Anderson & Gerbing, 1988).

The standardized factor loadings and construct reliability for the measurement model are presented in Table 1. The individual item loadings on the constructs were all highly significant ($p < .001$; t values > 21) with all loadings in excess of 0.6 and most greater than 0.8. The individual indicators have substantial variance attributed to the underlying latent construct (convergent validity). The value of the composite reliability coefficient ranged from 0.84 to 0.90, and was greater than the recommended 0.60 (Bagozzi & Yi, 1988).

Table 1. Standardized Confirmatory Factor Analysis Coefficients and Construct Reliability.

Constructs and Indicators	Standardized Factor Loadings	<i>t</i> Value	Composite Reliability
Satisfaction			0.90
Feeling satisfied	0.82	26.51	
Pleasant feeling	0.98	30.70	
Subjective ambivalence			0.84
Satisfied / dissatisfied	0.77	29.23	
Good / bad	0.87	33.64	
Delighted / boring	0.75	28.03	
Repurchase loyalty			0.84
Yearly frequency	0.60	21.60	
Recent frequency	0.87	33.71	
Intention to buy	0.89	34.53	

χ^2 (17 *df*) = 31.45; *p* = .0176; RMSEA = .027; NNFI = .99; CFI = 1.00.

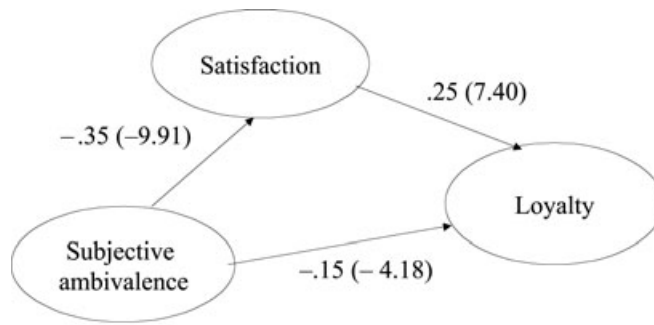
Table 2. Descriptive Statistics and Intercorrelations between Constructs Used in the Study.

	1	2	3	Mean	SD	Range
1. Satisfaction	1.00			5.89	1.20	1–7
2. Ambivalence	–.35	1.00		3.73	1.49	1–7
3. Loyalty	.31	–.24	1.00	4.17	1.17	1–8.1

Descriptive statistics and the correlations among the factors are displayed in Table 2. All correlations are below 0.36, suggesting discriminant validity. That is, each correlation is less than 1.00 by an amount exceeding twice its respective standard error (J. C. Anderson & Gerbing, 1988). To further assess the discriminant validity of the subset of measures, a procedure recommended by Bagozzi, Yi, and Philips (1991) was used. Within each subset of measures, pairs of constructs were examined in a series of two-factor confirmatory models. A chi-square difference test was conducted. The results suggest that for all pairs of constructs the two-factor solution was better ($p < .001$) than the single-factor solution. For example, combining satisfaction and ambivalence into a single factor produced a significantly worse fit ($\chi^2 = 767.13$; RMSEA = 0.357) than did a model treating these as two separate factors ($\chi^2 = 7.01$; RMSEA = 0.025). In sum, the measures of the proposed constructs achieve high reliability and convergent and discriminant validity.

The Impact of Ambivalence on Satisfaction and Loyalty

Structural-equation analysis was used to test the hypothesized relationships between ambivalence, satisfaction and loyalty. Figure 1 presents the full model with standardized path coefficients and *t* values in



Chi-Sq = 31.45, $df = 17$, $p = .018$
 RMSEA = .027, NNFI = .99, CFI = 1.00
 T-values are in parentheses.

Figure 1. Structural relationships with standardized parameters.

parentheses. The measures of goodness of fit for the entire model are good (RMSEA = 0.027, NNFI = 0.99, CFI = 1.00).

As expected, subjective ambivalence was negatively related to satisfaction (standardized beta = -0.35 , $p < .01$). That is, ambivalent consumers are in general less satisfied (and therefore less extreme) compared to unambivalent consumers. Ambivalence explained 13% of the variance in satisfaction. The correlation between ambivalence and loyalty was $-.24$ ($p < .01$). Even though most of this relationship is mediated through satisfaction, the results confirmed a significant negative relationship between ambivalence and loyalty (standardized beta = $-.15$, $p < .01$). In general, more ambivalent consumers are less loyal than those who exhibit less ambivalence. However, ambivalence did not explain more than a marginal amount of variation in loyalty. Allowing a direct path between ambivalence and loyalty increased R^2 from 10% to 11%.

Ambivalence as a Moderator of the Satisfaction–Loyalty Relationship

Because ambivalence has been shown to be a predictor of loyalty, using it to form subgroups to examine its role as a moderator is not appropriate (Peters & Champoux, 1979). Thus, moderated regression analysis (MRA) (Sharma et al., 1981) with latent variables (Jöreskog & Yang, 1996) was conducted.² MRA makes use of the estimates in three regression equations:

$$\eta = \gamma_1 \xi_1 + \zeta, \quad (1)$$

$$\eta = \gamma_1 \xi_1 + \gamma_2 \xi_2 + \zeta, \quad (2)$$

$$\eta = \gamma_1 \xi_1 + \gamma_2 \xi_2 + \gamma_3 \xi_1 \xi_2 + \zeta, \quad (3)$$

²The authors are grateful to one of the reviewers for suggesting this method to test interactions.

where η is loyalty, ξ_1 is satisfaction, ξ_2 is ambivalence, and $\xi_1\xi_2$ is the product (interaction) term. The pattern of significant coefficients across these equations indicate the presence and type of moderators.

Models were estimated with maximum likelihood (ML) and weighted least squares based on the augmented moment matrix (WLSA) by applying LISREL 8.52. The measures of goodness of fit for the ML-estimated models are good (RMSEA ranges from 0.035 to 0.057). For the WLSA-estimated model the fit was bad (RMSEA > 0.40). Even though WLSA is expected to give correct chi squares and consistent standard errors of parameter estimates (Jöreskog & Yang, 1996), a very large sample is necessary to accurately estimate the asymptotic covariance matrix needed for WLSA. As pointed out by Hu, Bentler, and Kano (1992) the WLS methods is reliable only for large sample sizes ($N > 5,000$ for relatively complex models). If the sample is not large ($N = 1,194$) it is probably best to use ML (Jöreskog & Yang, 1996). The pattern of significance reported in Table 3 suggests that ambivalence is not a moderator variable. Although the multivariate normality assumptions of maximum-likelihood estimation are not met, the interaction coefficient was not significant under either estimation procedure.

Table 3. Maximum Likelihood Estimates for the MRA Equations with Latent Variables.

γ_1 (satisfaction)	γ_2 (satisfaction)	γ_3 (sat*amb)	RMSEA	NNFI	CFI	R^2
.32 (9.29)*			0.057	0.98	.99	.10
.26 (7.36)*	-.16 (-4.34)*		0.035	0.99	1.00	.12
.17 (7.45)*	-.08 (4.36)*	-.002 (-0.11)	0.046	0.99	.99	.12

* Significant at $p < .05$, t values are in parentheses.

DISCUSSION AND IMPLICATIONS

As a cornerstone of the marketing concept, customer satisfaction has been embraced by practitioners and academics alike as the highest-order goal of the company (Oliver, 1997), and is believed to have great influence on customer loyalty (Fornell et al., 1996; Hennig Thurau & Klee, 1997). If some products or services produce a mixture of good and bad evaluations across different attributes, it could be hard to express a global evaluation without some kind of conflict, mixed feelings, or ambivalence. Standard satisfaction scales do not allow for the expression of evaluative ambivalence. Recent research within social psychology and related disciplines has provided evidence that ambivalence influences different facets of attitude evaluation (e.g., extremity) as well as the relationship between attitude and behavior (Conner & Sparks, 2002; Jonas et al., 2000a). The study reported here integrates these findings by investigating how ambivalence influences consumer satisfaction, loyalty, and the relationship between satisfaction and loyalty.

If plotted, the relationship between ambivalence and satisfaction would be expected to be an inverted U. The present findings, however, indicate a negative relationship between ambivalence and satisfaction (standardized beta = -0.35 , $p < .01$). As is often the case when measuring satisfaction, the majority of consumers in this study were satisfied with seafood, resulting in a highly skewed distribution. Thus, it is likely that the left part of the inverted U is missing (Olsen, 1999), because ambivalent consumers are less satisfied than unambivalent ones. This is in accordance with earlier findings on attitude extremity (Brömer, 1998; Lavine et al., 1998) and studies indicating that ambivalent respondents tend to use the midpoint responses of an attitude scale (DuBois & Burns, 1975; Klopfer & Madden, 1980).

Social psychologists have assumed ambivalence to be a pure moderator between attitude and behavior. Unfortunately, this assumption has ignored the possible relationship between ambivalence and the dependent variable (often behavioral intention). The correlation between ambivalence and loyalty was -0.24 ($p < .01$). The results suggest that even though a great deal of the variance in ambivalence is related to satisfaction, ambivalence has an independent and direct effect on repurchase loyalty or frequency of behavior (standardized beta = -0.15 , $p < .01$). Ambivalent consumers are not only less loyal because they are less satisfied, but they are less loyal for other reasons, as well.

One possible reason is that their evaluations of satisfaction are less stable than their reports of motivation and behavior (Jonas et al., 1997). Research indicates that weak attitudes are less related to behavior than strong attitudes (Krosnick & Petty, 1995), and there is some evidence that attitudes show lower temporal stability at higher levels of ambivalence (see Jonas et al., 2000a for a review). Lavine et al. (2000) found a negative relationship between attitude involvement and subjective ambivalence. Jonas et al. (2000b) found that when a moderating effect of ambivalence on attitude–behavior was found, it was accompanied by a parallel moderating effect of ambivalence on attitude stability, supporting the rationale that an effect of ambivalence on the satisfaction–loyalty relationship is mediated by lower temporal stability of highly ambivalent attitudes. It is reasonable to assume that ambivalent consumers are less involved in and do not think about the product or brand as frequently as nonambivalent consumers. This may cause temporal instability (Jonas et al., 2000b) and have a direct negative influence on satisfaction.

When the nature of interactions was examined, ambivalence was not found to moderate the satisfaction–loyalty relationship. This finding is consistent with Jonas et al. (2000b), but in conflict with others (Povey et al., 2001; Sparks et al., 2001). In that respect, the questions of how and when satisfaction behaves as a moderator (Jonas et al., 1997) need further research. Such investigations might consider the way constructs are measured (Brömer, 1998; Conner & Sparks, 2002; Jonas et al., 2000b), and perhaps how ambivalence is influenced by internal and external

variables such as knowledge, self-interest, norms, or involvement (Armitage & Conner, 2000; Sparks et al., 2001).

A moderation effect may be hard to detect if much of the variation in the dependent variable is already captured by main effects (Jonas et al., 2000b). As noted above, ambivalence was related to the measure of satisfaction and mediated through satisfaction onto loyalty. In addition, it had a separate direct effect on loyalty, with the same negative valence as its relationship to satisfaction. Its direct relationship to loyalty classifies ambivalence as an antecedent variable (as in these results) or as a quasimoderator (Sharma et al., 1981) in some other studies (Povey et al., 2001; Sparks et al., 2001). The current research extends traditional social psychological literature by recognizing the danger of using a median split to test for moderating effects of ambivalence given its role as an antecedent (Peters & Champoux, 1979). This study also used structural equation modeling (Jöreskog & Yang, 1996) to test the significance of the interaction effect.

Ambivalent attitudes are more likely to be changeable than attitudes showing no ambivalence, having implications for those interested in promoting behavioral change (Povey et al., 2001). For example, it is possible that consumers could be segmented on the basis of ambivalence and other strength-related variables. Ambivalent consumers would be targets for persuasive messages to increase the value of their positive beliefs and/or decrease their negative beliefs toward buying or using a particular product, brand, or service. Strategies for consumers with only negative beliefs might require longer-term efforts, first creating ambivalent attitudes and then converting them to positive at a later point.

Limitations and Future Research

Previous research into attitudinal ambivalence has operationalized ambivalence in different ways. The present effort used one direct measure of subjective ambivalence in this study. Future research should be extended to test the consequences of ambivalence on satisfaction and loyalty with different forms of both “subjective” (Conner & Sparks, 2002) and “objective” ambivalence (Breckler, 1994; Priester & Petty, 1996). Despite the use of a large nonstudent sample, the results are based on only one behavioral situation. Even though most studies within social psychology have used food behavior as their focus, future research should be extended to other products and services. Because services often tend to be more complex and involve more independent evaluations (Oliver, 1997), future research should examine how ambivalence interacts with service satisfaction and loyalty.

Like many studies in the area of loyalty and satisfaction (Peterson & Wilson, 1992), the Norwegian respondents reported being very satisfied with seafood. This of course results in a negatively skewed distribution. In studies where this is not the case (e.g., a positively skewed distribution with mostly dissatisfied customers), it might be expected that more

ambivalent consumers would be dissatisfied and less extreme in the domain of dissatisfaction (Brömer, 1998; Olsen, 1999). Under such circumstances a positive relationship between satisfaction and ambivalence would be expected. A weakness of the present study is that it is correlational. Future research might consider manipulating one or both antecedent variables to test the time-based nature of the relationship between the elements in the model, and use different measure of loyalty as the dependent variable.

Within established satisfaction and quality research (see Oliver, 1997, for a review), there are several evaluative propositions that may be affected by ambivalence and could cause conflict in expressions of evaluations. People may be ambivalent about expectations or they may be ambivalent about performance. They may, furthermore, be ambivalent about their evaluations or expectations of comparison objects, norms, or standards (e. g., what a service should offer, what an ideal product is, or the performance of a comparative product), or about their disconfirmation evaluation (better or worse than expected). At the very least, people may be ambivalent about the meaning of items due to semantics or the response format (e.g., what is *ideal*, *should be*, or *quality*?). To the authors' knowledge, no research has examined ambivalence as it relates to perceived quality, satisfaction, expectations, disconfirmation, or other constructs or items within consumer research.

It is worth mentioning once again that the measurement of loyalty involves past behavior that is predicted by independent variables measured in the present. Although temporal precedence is violated in a causal sense, the data themselves are cross sectional. Thus, it is only possible to conclude that the data are consistent with (based on model fit) the restrictions placed upon them by a causal diagram.

Recent research on consumer satisfaction (e.g., Fournier & Mick, 1999; Spreng & Page, 2001) and on the relationship between satisfaction and repurchase behavior (Holmburg & Giering, 2001; Mittal & Kamakura, 2001) suggests complicated relationships caused by differences in confidence, involvement, individual characteristics, social forces, or response bias. Future research is needed to investigate how these variables interact with ambivalence, satisfaction, and loyalty.

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